20/01/25 M/S

USN | | | | | | | | | | [23CSE102]

FIRST SEMESTER B.Tech EXAMINATION, JANUARY-2025

Object Oriented Programming

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		Questions	Blooms Level	CO's	Marks
		PART-A	**5 x2=10**		
1.	**a.**	Recall Software Engineering.	L 1	CO1	**02**
	b.	Infer the concept of Encapsulation.	L 2	CO1	**02**
	c.	Interpret Lambda Expressions in Java with suitable example.	L 2	CO2	**02**
	d.	Explain the concept of polymorphism in Java.	L 2	CO5	**02**
	e.	Define exception handling, and how does it help in managing runtime errors.	L 2	CO6	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Create a Java program to get input from the user and add three numbers.	L 3	CO1	**05**
		OR			
	b.	Explain the significance of SOLID principles in software design and how they enhance maintainability and scalability.	L 2	CO1	**05**
3.	**a.**	Experiment with Getters and Setters concept in Java.	L 3	CO2	**05**
		OR			
	b.	Infer the concept of packages in Java and explain their advantages with appropriate syntax.	L 2	CO2	**05**
4.	**a.**	Analyze the role of super() in Java and explain how it is used to access members of a superclass with an example program.	L 3	CO3	**05**
		OR			
	b.	Apply the concept of interfaces in Java by creating an example program that demonstrates the implementation of multiple interfaces.	L 3	CO3	**05**
5.	**a.**	Demonstrate the concept of polymorphic references in Java with an example program.	L 3	CO5	**05**

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		OR			
	b.	Analyze the difference between "low coupling and high cohesion" versus "high coupling and low cohesion". Which one is considered a better design principle, and explain the reasoning behind it.	L 3	CO5	**05**
6.	**a.**	Explain the following concepts in Java: **i.** Syntax Error **ii.** Errors **iii.** Exceptions	L 2	CO6	**05**
		OR			
	b.	Explain the difference between checked and unchecked exceptions in Java.	L 2	CO6	**05**
7.	**a.**	Criticize abstract methods and abstract classes in Java with suitable program.	L 3	CO4	**05**
		OR			
	b.	The final class cannot be inherited. Justify this statement with an example program.	L 3	CO4	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Summarize the core Object-Oriented Programming (OOP) concepts in Java with suitable examples for each concept.	L 2	CO1	**10**
		OR			
	b.	Inspect static variable and method in Java with an example program.	L 3	CO1	**10**
9.	**a.**	Examine the concept of the access modifiers in Java with suitable program.	L 2	CO2	**10**
		OR			
	b.	Illustrate the concept of Constructor in Java with an example.	L 2	CO2	**10**
10.	**a.**	Explain Inheritance in Java with its types and apply single inheritance concept with an example program.	L 3	CO3	**10**
		OR			
	b.	How can we achieve multiple inheritance in JAVA? Explain with an example program.	L 3	CO3	**10**
11.	**a.**	Illustrate the concept of polymorphism in Java, with an example program for dynamic polymorphism.	L 3	CO5	**10**
		OR			

	b.	Design a Java program to illustrate the concept of static polymorphism by implementing the following: **i.** A method to add two integers. **ii.** A method to add three integers. **iii.** A method to add two floating-point numbers.	L 3	CO5	**10**
12.	**a.**	Explain Exception Handling Keywords in Java, with an example program.	L 3	CO6	**10**
		OR			
	b.	Create a user-defined exception in Java. Explain the steps involved in defining a custom exception class and how to use it in a program.	L 3	CO6	**10**
13.	**a.**	Criticize the concepts of Coupling and Cohesion with an example program demonstrating low coupling.	L 3	CO5	**10**
		OR			
	b.	Explain the following with an example program: i. Method overloading ii. Method overriding	L 3	CO5	**10**